Enrico Didoli

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Applied Mathematics & Machine Learning

Profile

Applied Mathematician and Data Scientist with a strong probabilistic background, focused on statistical learning for high-dimensional biomedical data. I build deep learning pipelines for critical domains where mistakes carry real consequences, correcting for severe class imbalance to reach reliable performance. My core research interest is Bayesian methods and uncertainty-aware model calibration, building models that stay interpretable and trustworthy when the underlying data is limited or noisy.

Education

2023-2026 **M.Sc. in Stochastics and Data Science**

University of Turin, Turin (Italy) | **Final Grade**: 105/110

- **Thesis**: "Detecting Measurable Residual Disease in B-cell Acute Lymphoblastic Leukaemia using Convolutional Neural Networks"
- **Supervisors**: Prof. Yvan Saeys, Prof. Francesca Romana Crucinio
- **Links**: Thesis Repository | Thesis PDF

2024 **Erasmus+ Exchange Program**

Ghent University, Belgium (6 months study period)

2020-2023 **B.Sc. in Economics and Data Science**

University of Turin, Turin (Italy) | **Final Grade**: 102/110

- **Thesis**: "Total variation between the law of the number of fixed points in a random permutation and its limit Poisson distribution"

Research Experience

2025–2026 **M.Sc. Thesis Researcher**

Department of Mathematics, University of Turin & University of Ghent

Turin, Italy - Ghent, Belgium

- Extended and trained the CellCNN framework for Multiparameter Flow Cytometry sample classification using Tensorflow.
- Designed patient-level data augmentation to prevent data leakage and improve generalization.
- Tuned classification thresholds and employed Bayesian optimization to select internal CellCNN data-augmentation parameters on clinical datasets.
- **Key Results**: Recall = 0.95, F1 = 0.947 on held-out test set, with fully reproducible pipeline.

Projects

2024 **Inverse Problems: Determination of Time-Dependent Heat Source in the Heat Equation**, *Group project, University of Ghent*

- Reformulated an inverse source problem for the heat equation as a variational (weak) problem under Neumann and Dirichlet boundary conditions.
- Proved existence and uniqueness of the semi-discretized problem via Rothe's method and the Lax-Milgram lemma.
- Derived a priori energy estimates and proved convergence to the weak solution using Kačur's lemma and Grönwall's inequality, including an error bound for the time discretization.

2024 **Calibration of ML Models Across Domains**, *University of Ghent*

- Compared Temperature Scaling and Isotonic Regression for model calibration on CIFAR-10, part of coursework during my Erasmus+ semester.
- Found that calibration effectiveness is model- and domain-specific, giving me direct hands-on experience quantifying and validating model uncertainty.
- Link: Project Repository

2026 **Independent project (in progress): Single-Cell Classification**

- CellCNN filter analysis for single-cell classification (Extension of Master's Thesis research).
- Clustering and filter selection: comparing Logistic Regression and mean function across filter responses to determine single-cell classification thresholds.

2026 **Independent project (in progress): Retrieval Augmented Generation (RAG)**

- Bayesian-optimization pipeline to tune chunking, overlap, and generation parameters in a RAG system; found response quality bottlenecked by base model capacity rather than pipeline configuration.

Professional Background

2018-2021 Pastry Chef and Chocolate Artisan (Luxury Hospitality, Italy)

- Worked in high-pressure, precision-demanding environments before transitioning to academia to pursue applied mathematics. Cultivated adaptability, resilience, and strict attention to detail.

Skills

Deep Learning	Computer Vision (CNNs, Transfer Learning), Foundation Models (Qwen), TensorFlow/Keras
Machine Learning	Scikit-learn, Optuna (Bayesian Hyperparameter Optimization), Imbalanced-learn
Data Science & Analysis	Pandas (DataFrames), Numpy, flow cytometry analysis, handling clinical dataset variability
Programming & Tools	Python, Jupyter, GitHub, LaTeX, SQL (Basic), R (Basic)
Soft Skills	Teamwork, problem-solving, communication, creative thinking.
Languages	English (B2/C1), Italian (native).

References

- Prof. **Francesca Romana Crucinio** | University of Turin, Torino (Italy)

  Professor, Dept. of Economics, Social Studies, Applied Mathematics and Statistics

  Email: francescaromana.crucinio@unito.it

- Prof. **Yvan Saeys** | University of Ghent, Ghent (Belgium)

  Professor, Computer Science

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- Prof. **Pierpaolo De Blasi** | University of Turin, Torino (Italy)

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